Experiment 1, non-visual arms: Why the Estimators Err the Way They Do

Insight analysis for Chapter 4, Section 4.1

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Purpose of this document. The raw write-up of Experiment 1 (`insights/experiment_1/`) reports how far each estimator is from the truth on the four realistic-warehouse recordings. This document asks what kind of error each one makes and when it is made: whether wheel odometry's error is a scale or a heading error, why IMU dead reckoning diverges, when and why the EKF's covariance stops describing its error, what the transferred noise model does to that filter, and what the visual estimators' reported error consists of once they are scored at the IMU point. Each question is tested within every recording, so the recording is its own control, and the answer carries the evidence for and against it. Every number is written by `script/exp1_insights.py` into `output/compare/simulation/exp1-insights/`; the tables are input from there unchanged.

1 Questions and method

Six questions were put to the four `dataset_allsensor_realistic` recordings, one run per estimator per recording: `allsensor_realistic_000` and `allsensor_realistic_002` on the U layout, `allsensor_realistic_001` and `allsensor_realistic_003` on the I layout, with the two odd-numbered / even-numbered pairs splitting static from moving traffic, as the chapter's simulation subsection records. The routes run 103–172 m.

- Q1. Is wheel odometry's error a distance (scale) error, a heading error, or heading error carried into position?
- Q2. Does each estimator's error grow faster while the robot turns than on straights?
- Q3. Does IMU dead reckoning diverge through its heading or through its velocity?
- Q4. When does the EKF's covariance stop describing its error?
- Q5. What does the transferred noise model (the VSLAM file's IMU constants) do to the EKF, mechanically?
- Q6. For the visual estimators, scored at the IMU point: what is their reported error made of, does it grow at turns or with few features, and what was the one failure?

Baselines. The wheel, IMU and EKF estimates start from the ground-truth pose and are compared to the raw model-origin ground truth, as in the raw write-up. The error vector is rotated into the ground-truth path frame, giving an along-track part (distance travelled) and a cross-track part (lateral), and the heading error is the estimate's yaw against the truth's, zeroed at the start. To separate heading from velocity error, the true velocity is integrated through the heading error alone; the ratio of that prediction to the actual error says how much of the error the heading explains. Growth rates while turning (ground-truth yaw rate above $10\,^\circ\,\mathrm{s}^{-1}$) and on straights are the median growth of the error within the segments of each kind, the convention the translation and rotation error patterns of `output/compare/simulation/exp1-`

`error-pattern/` also use. NEES is computed from the logged covariance at 1 Hz; the filter "leaves its band" at the first time it stays above the 97.5 % value of a 3-DOF χ^2 for 5 s.

The two EKF arms. `output_ekf_measured` is the filter driven by the noise constants measured on this rig and is reported in the chapter as the fused baseline; `output_ekf` is the same filter driven by the IMU constants carried in the VSLAM configuration file, kept as a diagnostic. The tree names are the reverse of what they suggest and are easy to swap.

Visual estimators. VINS-Fusion and ORB-SLAM3 report the IMU, 0.338 m ahead of the model origin, so they are scored against the ground truth moved to that point. The initialisation transient is cut and the start-anchored error is taken over the tracking regime — before the first impossible-speed jump and below 3 m of error — with the same turning contrast as the baselines. Whole-run ATE is reported both in three dimensions, as the chapter does, and in the plane.

2 Insight 1: wheel odometry's error is heading error made in the corners

Wheel odometry measures distance travelled almost exactly and loses its heading, and its position error is that heading error integrated along the route. Table I.1, table I.2 and figure I.1 carry the evidence.

Table I.1: Anchored error of the proprioceptive baselines in the ground-truth path frame: along-track (distance) and cross-track (heading) RMS and final values, the heading error itself, the path-length (scale) error of the estimate, and the share of the final position error that the heading error alone reproduces when the true velocity is integrated through it. Model-origin reference. Source: `output/compare/simulation/exp1-insights/decomposition.csv`.

Recording	Estimator	RMS [m]	along RMS [m]	cross RMS [m]	final [m]	heading RMS [°]	scale [%]	heading-only share of final
`allsensor_realistic_000`	Wheel odometry	5.89	4.69	3.56	5.66	18.05	0.0	1.04
	IMU dead reckoning	36.19	18.46	31.12	76.01	0.84	53.8	0.01
	EKF @ measured noise	0.33	0.24	0.22	0.35	0.44	0.5	1.03
	EKF @ VSLAM noise	2.49	2.17	1.21	3.42	10.63	13.7	0.98
`allsensor_realistic_001`	Wheel odometry	1.40	1.01	0.97	2.90	7.27	-0.4	1.02
	IMU dead reckoning	15.40	8.57	12.80	39.82	0.55	-11.9	0.01
	EKF @ measured noise	0.26	0.25	0.08	0.26	0.37	-0.3	0.44
	EKF @ VSLAM noise	0.94	0.75	0.58	0.83	4.51	11.1	2.30
`allsensor_realistic_002`	Wheel odometry	6.54	5.04	4.17	6.08	25.00	0.1	1.01
	IMU dead reckoning	25.88	13.51	22.07	39.17	1.91	41.3	0.01
	EKF @ measured noise	0.59	0.49	0.33	0.30	1.04	0.4	0.68
	EKF @ VSLAM noise	2.06	1.77	1.05	3.29	13.65	10.5	1.05
`allsensor_realistic_003`	Wheel odometry	6.71	5.70	3.54	7.19	35.82	-0.1	1.02
	IMU dead reckoning	13.72	7.04	11.78	35.47	1.68	-37.3	0.01
	EKF @ measured noise	0.79	0.71	0.35	1.49	5.72	0.4	1.13
	EKF @ VSLAM noise	4.44	3.84	2.24	6.23	28.01	8.2	1.08

The path length the wheels report is within 0.4 % of the true one on every recording, while the heading error reaches 7.3–35.8° RMS. Integrating the true velocity through that heading error alone, with the true speed, reproduces the final position error to within 4 % on all four recordings (heading-only share 1.01–1.04). Position error and heading error are therefore not two findings but one: there is no scale error to speak of, and what the chapter reports as 1.40–6.71 m of position error is 7–36° of yaw converted into displacement by the length of the route.

The error is made in the corners and paid for on the straights. Table I.2 separates the two rates: the heading error grows at 1024 m° s^{-1} while the ground-truth yaw rate exceeds 10 ° s^{-1} against 13 m° s^{-1} on straights, a factor of 77, and it is the larger of the two in three of the four recordings. The position error

Table I.2: Growth rate of the position error and of the heading error while the ground-truth yaw rate exceeds $10\,^\circ\,s^{-1}$ and on straights, median over the four recordings, with the number of recordings in which the turning rate is the larger ($n_>$). Baselines: anchored error over the whole run. Visual estimators: start-anchored error at the IMU point, tracking regime only (before the first jump, below 3 m); heading contrast not computed for them.

Estimator	n	position [mm s^{-1}]			heading [m° s^{-1}]		
		turning	straight	$n_>$	turning	straight	$n_>$
Wheel odometry	4	21.5	69.0	0	1024.5	13.3	3
IMU dead reckoning	4	122.0	541.0	1	19.6	21.3	2
EKF @ measured noise	4	12.1	1.9	3	10.0	17.4	2
EKF @ VSLAM noise	4	110.3	8.4	3	659.3	56.0	3
VINS-Fusion	4	0.2	6.2	1	—	—	—
ORB-SLAM3	4	-1.8	1.4	1	—	—	—
ORB-SLAM3 (no LC)	4	0.6	0.4	1	—	—	—

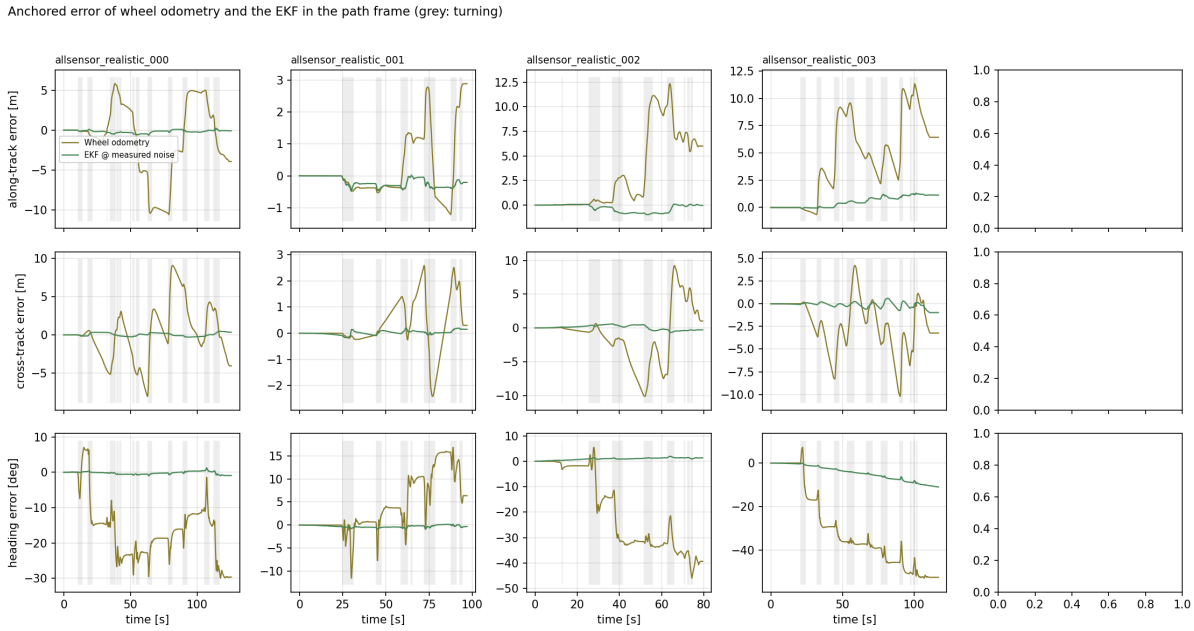


Figure I.1: Anchored error of wheel odometry and the EKF on each recording, in the ground-truth path frame: along-track error, cross-track error and heading error against time. Grey bands are the moments the ground-truth yaw rate exceeds $10\,^\circ\,s^{-1}$.

does the opposite — 21.5 against 69.0 mm s^{-1}, larger on straights in all four — because a heading error acquired in a corner only becomes displacement over the straight that follows it. A reader who looks only at position growth would conclude that turning is harmless to wheel odometry, which is the wrong way round.

Against the insight. The weakest case is `allsensor_realistic_001`, where 7.3° of heading error gives 1.40 m RMS, a fifth of the error on the other three; its route is also the shortest. The mechanism is the same there, the lever is just shorter. Nothing in these data separates slip at the wheel from a wheel-radius or track-width error in the kinematic model, both of which would present as heading error in a corner.

3 Insight 2: IMU dead reckoning diverges through its velocity, not its heading

IMU-only dead reckoning keeps its heading to within a few degrees and loses its position through the velocity integral. Table I.3 and figure I.2 carry the evidence.

Table I.3: IMU dead reckoning: exponent k of a power-law fit $|e| \propto t^k$ to the anchored position error after the first 5 s (with its R^2), the final heading error, and the share of the final position error that integrating the true velocity through the heading error alone reproduces. A share near one means the position error is heading error integrated along the path; a share well below one means the velocity (accelerometer) integration itself is wrong.

Recording	k	R^2	final error [m]	final heading [°]	heading-only share
`allsensor_realistic_000`	1.53	0.80	76.0	1.8	0.01
`allsensor_realistic_001`	2.22	0.82	39.8	-0.8	0.01
`allsensor_realistic_002`	1.71	0.92	39.2	3.3	0.01
`allsensor_realistic_003`	0.91	0.85	35.5	-3.2	0.01

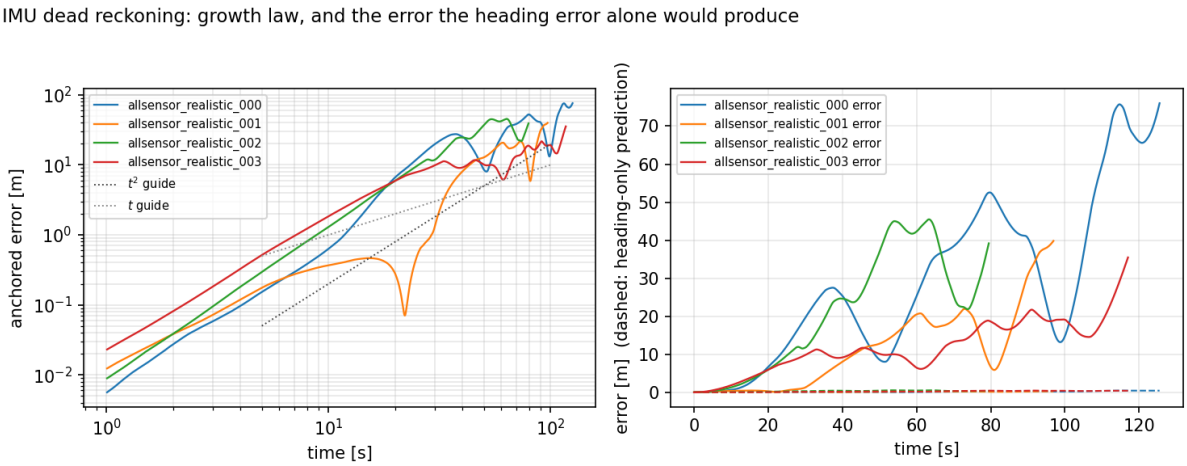


Figure I.2: IMU dead reckoning on the four recordings. Left: anchored position error against time on logarithmic axes, with t and t^2 guides. Right: the same error on linear axes with, dashed, the error that the heading error alone would produce.

The heading error of the IMU integration is 0.55–1.91° RMS and 3.3° at worst at the end of a run — an order of magnitude better than wheel odometry's — yet the final position error is 35.5–76.0 m on routes of 103–172 m. The heading-only prediction accounts for 1 % of it on every recording. The gyroscope is not the problem; the accelerometer channel is. The path length the integration reports is wrong by

12–54 %, and the sign of that error is not even consistent across the four recordings (53.8, −11.9, 41.3 and −37.3 %), which is what a bias of either sign integrated twice looks like.

The growth law is consistent with that reading. A power-law fit after the first 5 s gives $k = 0.91 - -2.22$ with R^2 between 0.80 and 0.92: between one and two integrations, as a velocity error that is partly random walk and partly bias should be. The turning contrast in table I.2 agrees — position error grows at 122 against 541 mm s^{-1}, faster on the straights, and faster while turning in only one of the four recordings. The divergence accumulates wherever the robot is moving, not where it manoeuvres.

Against the insight. k is not the same on all four and `allsensor_realistic_003` fits a sub-linear 0.91, so "diverges as $t^{1.5}$" would overstate what four runs support; the defensible claim is that the error grows faster than the distance travelled and that the heading explains almost none of it. No bias estimate was logged, so the split between bias and random walk is inferred from the exponent rather than measured.

4 Insight 3: the EKF removes the heading error, and its covariance stops describing its error at the first turn

Fusing the gyroscope with the wheels removes the heading error that Insight 1 isolated, and takes the position error down by between a factor of 5 and a factor of 18. The filter's own covariance stops covering that error at the first corner of the route. Table I.1, table I.4 and figure I.3 carry the evidence.

Table I.4: When the EKF's 3-DOF NEES first stays above the 97.5 % band (9.35) for 5 s, as time since the start and as distance travelled, against the time of the first ground-truth heading change (yaw rate above 10 ° s^{-1}); for the EKF with its measured per-recording noise model and for the same filter on the VSLAM noise file. NEES sampled at 1 Hz from the logged covariance. "never": the band is not left for 5 s in the run.

Recording	first turn	EKF @ measured noise			EKF @ VSLAM noise		
	[s]	leaves at [s]	at [m]	median NEES	leaves at [s]	at [m]	median NEES
`allsensor_realistic_000`	10.5	10.0	11.9	57.5	10.0	11.9	15.0
`allsensor_realistic_001`	24.7	25.0	33.8	34.2	27.0	36.0	3.6
`allsensor_realistic_002`	12.7	9.0	11.4	116.6	27.0	37.4	34.0
`allsensor_realistic_003`	20.6	20.0	28.2	73.7	21.0	28.7	139.2

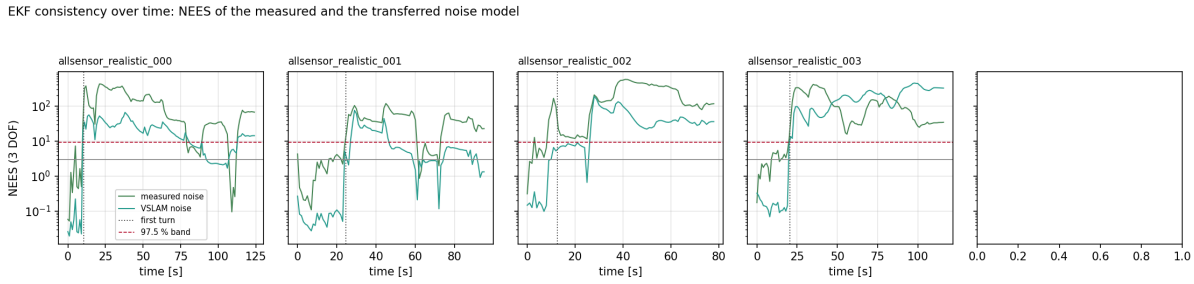


Figure I.3: 3-DOF NEES of the EKF against time on each recording, for the measured per-recording noise model and for the VSLAM noise file, with the first ground-truth heading change (dotted), the 97.5 % band (dashed) and the consistent value of 3 (solid).

With the measured noise model the filter holds 0.26–0.79 m RMS against wheel odometry's 1.40–6.71 m, its heading error falls to 0.37–5.72°, and its path-length error stays within 0.5 %. The accuracy claim of the chapter's simulation subsection is, mechanically, a heading claim: the wheels were already measuring distance correctly, and the gyroscope supplies the one quantity they got wrong.

The covariance tells a different story. The 97.5 % band of a 3-DOF χ^2 is 9.35; the filter's median NEES over a run is 34.2–116.6, between four and twelve times it. It leaves the band at 10.0, 25.0, 9.0

and 20.0 s against first ground-truth heading changes at 10.5, 24.7, 12.7 and 20.6 s — within 4 s of the first turn on all four recordings, and on `allsensor_realistic_002` slightly before it, as the robot begins to rotate. The filter is accurate and overconfident, and the overconfidence begins at the first heading change rather than accumulating with distance.

Against the insight. The residual error is not purely heading: the heading-only share of the final error is 0.44–1.13 across the four recordings, against wheel odometry's uniform ≈ 1.0, so part of what is left after fusion is something else. NEES is computed from the covariance the filter publishes, which omits any error in the measured noise constants themselves, and a single run per recording cannot distinguish an under-modelled turn from one unlucky corner.

5 Insight 4: the transferred noise model turns the EKF back into wheel odometry

Driving the same filter with the IMU constants carried in the VSLAM configuration file costs between a factor of 3.5 and a factor of 7.5 in accuracy, and the error it is left with has the signature of Insight 1. Table I.1 and table I.4 carry the evidence.

On the four recordings the transferred model gives 0.94, 2.49, 2.06 and 4.44 m RMS against the measured model's 0.26, 0.33, 0.59 and 0.79 m. Three things change together. The heading error returns, to 4.51–28.01° from 0.37–5.72°. A path-length error appears where the measured filter had none — 8.2–13.7 % against 0.5 % — so the filter is now mis-measuring distance as well as heading, which neither of its two inputs does on its own. And the error becomes a cornering error: in table I.2 its position error grows at 110.3 against 8.4 mm s^{-1} and its heading error at 659 against 56 m° s^{-1}, both larger while turning in three of the four recordings, where the measured filter's heading error is not.

The covariance moves the other way. The transferred model's median NEES is 3.6–139.2, and on `allsensor_realistic_001` the median 3.6 sits inside the band that the measured filter leaves at 25 s. Inflating the assumed IMU noise buys nominal consistency on one recording while losing a factor of 3.6 in accuracy on it: the filter is correctly uncertain about an estimate that is worse. A consistency check alone would prefer the wrong configuration.

Against the insight. Which constant does the damage is not isolated here — the VSLAM file supplies accelerometer and gyroscope noise and random-walk terms together, and no single-parameter sweep was run. The transferred model is not a competing design either; it is a configuration mistake that was made and measured, and it is reported as a diagnostic for that reason.

6 Insight 5: most of the visual estimators' reported error is vertical, and their one failure is a loop-closure event

Scored in the plane the visual estimators are better than the chapter's three-dimensional ATE says, because much of that ATE is height drift on a robot that cannot change height; their planar error does not grow at turns or with fewer features in any consistent way; and the one gross failure is not drift at all but a false loop closure. Table I.5, table I.6 and figure I.4 carry the evidence.

In four of the twelve runs the three-dimensional ATE is at least 2.5 times the planar one, and in two of those more than eight times. ORB-SLAM3 on `allsensor_realistic_003` is 0.49 m in three dimensions and 0.06 m in the plane, with a vertical RMS of 0.49 m; the same recording without the loop closer is 1.83 m against 0.13 m (vertical 1.82 m); and VINS-Fusion on `allsensor_realistic_002` is 0.66 m against 0.25 m (vertical 0.61 m). The robot never leaves the floor, so for these runs the reported

Table I.5: Visual estimators scored at the IMU point after the initialisation transient: whole-run aligned ATE in 3-D (as the chapter reports it) and in the plane, the vertical RMS error that separates them, length of the tracking regime, drift rate while turning and on straights in that regime, Spearman ρ between the matched-feature count (ORB-SLAM3 map inliers, VINS-Fusion tracked features) and the drift rate in 2 s bins, and the number of impossible-speed jumps of the estimate.

Recording	Estimator	ATE 3-D [m]	ATE planar [m]	vertical RMS [m]	regime [s]	turning [mm s^{-1}]	straight	ρ (feat., drift) (bins)	jumps
`allsensor_realistic_000`	VINS-Fusion	0.29	0.26	0.13	113	10.4	6.1	-0.25 (56)	0
	ORB-SLAM3	0.16	0.16	0.02	110	-4.9	7.7	0.04 (55)	3
	ORB-SLAM3 (no LC)	0.33	0.33	0.06	108	-0.4	0.1	0.04 (53)	3
`allsensor_realistic_001`	VINS-Fusion	0.22	0.13	0.18	70	-1.1	8.0	-0.06 (34)	0
	ORB-SLAM3	4.85	4.85	0.03	46	0.7	1.0	0.32 (23)	7
	ORB-SLAM3 (no LC)	0.27	0.06	0.27	70	1.6	5.5	-0.08 (34)	2
`allsensor_realistic_002`	VINS-Fusion	0.66	0.25	0.61	52	-15.1	-3.3	0.25 (25)	0
	ORB-SLAM3	0.28	0.28	0.06	52	3.3	-0.9	-0.08 (25)	4
	ORB-SLAM3 (no LC)	0.13	0.10	0.09	52	2.2	0.3	-0.22 (25)	1
`allsensor_realistic_003`	VINS-Fusion	0.27	0.19	0.18	94	1.4	6.2	-0.07 (46)	0
	ORB-SLAM3	0.49	0.06	0.49	94	-4.3	1.8	0.06 (46)	3
	ORB-SLAM3 (no LC)	1.83	0.13	1.82	94	-1.8	0.5	-0.15 (46)	3

Table I.6: Anatomy of the ORB-SLAM3 (loop closer on) run on `allsensor_realistic_001`: whole-run aligned ATE, the number and sizes of impossible-speed jumps, when they occur, how much of the net error change is taken in jump steps against ordinary drift, and the tracking-state losses the front end logged.

Aligned ATE [m]	4.78
Jumps	7
Jump times [s]	2.5, 2.6, 71.1, 71.3, 71.5, 72.0, 74.8
Jump sizes [m]	3.24, 0.58, 17.68, 4.29, 1.29, 0.77, 3.16
Net error change in jump steps [m]	4.56
Net error change in ordinary steps [m]	2.42
Frames not tracking OK	0
Route progress at the largest jump	0.78
Time of the largest jump [s]	71.1

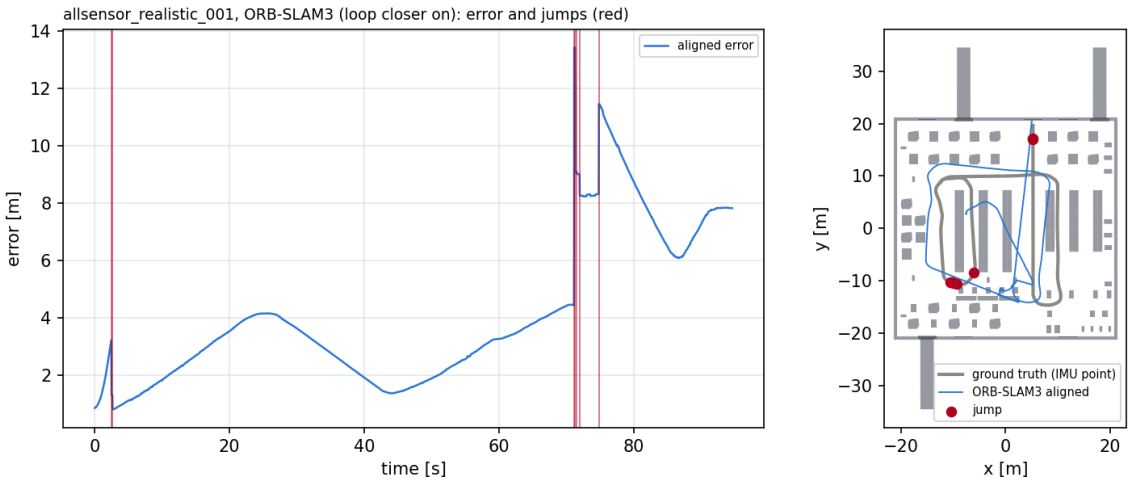


Figure I.4: ORB-SLAM3 with the loop closer on the I-layout static recording. Left: whole-run aligned planar error against time, with the impossible-speed jumps of the estimate in red. Right: the aligned estimate against the ground truth at the IMU point on the warehouse plan, with the jump locations.

figure is almost entirely an error in a degree of freedom the platform does not have, and it changes the ranking: on `allsensor_realistic_003`, ORB-SLAM3 loses to VINS-Fusion in three dimensions (0.49 against 0.27 m) and beats it threefold in the plane (0.06 against 0.19 m).

This is not uniform across the arms, and that is itself the finding. With its loop closer on, ORB-SLAM3 is flat — vertical RMS 0.02–0.06 m, three-dimensional and planar ATE equal to two decimals on three of the four recordings — while the same front end with the closer off carries up to 1.82 m of height error and VINS-Fusion up to 0.61 m. Height drift is what the loop closer removes when it fires on a correct match, which is the other half of the closure story below.

ORB-SLAM3 with its closer on `allsensor_realistic_001` is the exception to every planar figure above, at 4.85 m, and it is not drift. Table I.6 anatomises it: seven impossible-speed jumps, of which two are the initialisation transient at 2.5 s and five fall between 71.1 and 74.8 s at 78 % of the route; the largest moves the estimate 17.7 m in one step; 4.56 m of the net error change is taken in jump steps against 2.42 m of ordinary drift; and the front end logged no frames outside tracking-OK state. The estimator did not lose the robot — it confidently put it somewhere else. The same recording with the closer off tracks to 0.06 m in the plane. This is the false closure between the I layout's repeated aisles that the chapter's loop-closer analysis reaches independently from each run's `loop_closing.csv`, and it is the single reason ORB-SLAM3's accuracy table has an outlier on that recording.

Turning and feature count explain nothing here. The visual arms' drift rates while turning and on straights are 0.2 against 6.2, −1.8 against 1.4 and 0.6 against 0.4 mm s^{-1}, each larger while turning in one of the four recordings — a coin toss. Spearman ρ between matched-feature count and drift rate over 23–56 two-second bins runs from −0.25 to 0.32, with no consistent sign within or across arms.

Against the insight. The planar figure is the right one for comparing floor robots but it is not a licence to restate the chapter's numbers: the three-dimensional ATE is what the raw write-up reports and what the chapter must keep reporting, with the planar value beside it. And ORB-SLAM3 is not run-to-run repeatable — repeated runs of one configuration spread by up to 0.277 m, 80 % of their mean — so none of the sub-0.25 m differences quoted in this section, in either projection, is an effect.

7 What these data cannot answer

- **Four recordings, one run per estimator.** Every cell of table I.1 and table I.5 is a single run. The only repeated runs in the simulation campaign are the eighteen on the two mapping recordings, scored for accuracy into `output/compare/simulation/exp3-repeats/`, which bound run-to-run noise for the visual arms and not for the baselines. A difference between two cells here is a candidate, not an effect.
- **Mechanism is inferred from decomposition, not from an intervention.** Nothing was changed and re-run: no wheel-radius sweep, no bias injection, no single-parameter walk of the VSLAM noise file. The claims are about what each error is made of, which the decomposition measures, and not about what would happen if the suspected cause were removed.
- **Simulation only.** These are the four realistic-warehouse recordings. The chapter's real-robot subsection has no wheel, IMU or EKF baseline of comparable provenance, so none of the four baseline insights is shown to transfer to the rig.
- **Ground truth is the model origin for the baselines and the IMU point for the visual arms.** The two are 0.338 m apart. The baseline and visual numbers in this document are therefore not directly comparable to each other at the decimetre level.
- **No traffic contrast was tested.** Two of the four recordings carry moving traffic and two do not, but with one run each the pairing cannot separate a traffic effect from a layout effect or from run-to-run noise.

8 Summary

1. Wheel odometry has effectively no scale error (within 0.4 %) and 7.3–35.8° of heading error, which reproduces its whole position error to within 4 % on all four recordings. The heading error is made in the corners (77 times the straight-line rate) and converted into displacement on the straights that follow.
2. IMU dead reckoning keeps its heading to 0.55–1.91° and still reaches 35.5–76.0 m: the heading explains 1 % of the error, the path length is wrong by 12–54 % in either direction, and the error grows as t^k with $k = 0.91 - -2.22$.
3. The EKF with its measured noise model reaches 0.26–0.79 m by removing exactly the quantity wheel odometry got wrong, and its covariance leaves the 97.5 % band within 4 s of the first turn on every recording, running at four to twelve times the band thereafter.
4. The transferred VSLAM noise model costs a factor of 3.5–7.5 in accuracy, returns the heading error, introduces a 8.2–13.7 % path-length error that neither input has, and becomes a cornering error — while on one recording looking more consistent than the correct configuration.
5. For the visual arms, four of twelve runs report a three-dimensional ATE at least 2.5 times their planar error, almost all of it height drift; ORB-SLAM3 with its closer on is the one arm that is flat; and the single 4.85 m outlier is five jumps in 4 s from a false closure, not drift, on a recording the same front end tracks to 0.06 m with the closer off.

Consequences for Chapter 4. State which ATE a floor robot is being judged on, and report the planar value beside the three-dimensional one for every visual run. Attribute the EKF baseline's accuracy to heading observability rather than to fusion in general. Keep the transferred noise model in the chapter as a configuration diagnostic, with its consistency result, because it is the one place where a covariance check prefers the worse estimate. And label the 4.85 m ORB-SLAM3 cell as a closure failure in the table itself, so it is not read as drift.